

Project Title

Your Name

Abstract

This is a one-paragraph abstract of the review you plan to do in this project. It should provide some brief descriptions on the field and types of methods you will cover. Your project will be evaluated by its clarity, depth, and sufficiency in inclusivity of the works in the chosen topic.

1 Introduction

This section provides a concise introduction of the area to readers. It should explain the main problem to a general audience with reasonable background, such as your fellow graduate students.

2 Background

This provides some necessary background mathematics and formulas used in the area. It is to prepare readers for the methods to be reviewed.

3 Review of Methods

This section provides the main part of your review. It is better to review the methods class by class (or type by type). Group each type of methods in one subsection.

3.1 Method of Type I

Use subsections to review methods of the same or similar type. You should use some formulae to explain the key math behind and explicitly explain the difference between the methods within each class and/or between different classes.

3.2 Method of Type II

Another subsection.

4 Summary

A brief summary of what you have done in this review. You may also provide some prospective in the field.

5 Remarks

5.1 Academic Honesty

It is strictly prohibited to use any AI generative models, such as but not limited to ChatGPT, to generate any parts of your project. Your projects will be reviewed and evaluated by the latest Turnitin parasitism detector on whether the use of ChatGPT or copying from existing papers presents in your project. In such cases, your project will be scored 0, and you will get an F for your final grade. Additional Academic Dishonesty statements can be found in the syllabus.

5.2 Some Examples for References

You may find many review/survey papers of certain areas online to learn how to structure a review paper. Here are some examples: generative models [1], normalizing flows [2], and graph neural networks [3]. They are very comprehensive and lengthy as they cover a large field, but you can review on a smaller field in this project.

References

- [1] S. Bond-Taylor, A. Leach, Y. Long, and C. G. Willcocks. Deep generative modelling: A comparative review of vaes, gans, normalizing flows, energy-based and autoregressive models. *IEEE transactions on pattern analysis and machine intelligence*, 2021.
- [2] I. Kobyzev, S. J. Prince, and M. A. Brubaker. Normalizing flows: An introduction and review of current methods. *IEEE transactions on pattern analysis and machine intelligence*, 43(11):3964–3979, 2020.
- [3] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip. A comprehensive survey on graph neural networks. *IEEE transactions on neural networks and learning systems*, 32(1):4–24, 2020.